

Experiment 'p_5_1_b_average' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_5_1_b_average

Natural language statement: Write a method that returns the average of the arguments.

Method signature: p_5_1_b_average(x: real, y: real, z: real) returns (average: real)

Ensures

- `average == (x + y + z) / 3.0`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that returns the average of the arguments.

The signature should be:

`p_5_1_b_average(x: real, y: real, z: real) returns (average: real)`

The method should respect the following contract:

`ensures average == (x + y + z) / 3.0`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmyKqB8bEoOw6W3BvBA0lFkSNAB3s

Model: gpt-5.5-2026-04-23

Created at: 1780564044

Finish reason: stop

Usage: 282 tokens in, and 63 tokens out

```
{  
  average := (x + y + z) / 3.0;  
}
```

Final Program

```
method p_5_1_b_average(x: real, y: real, z: real) returns (average: real)  
  ensures average == (x + y + z) / 3.0  
{  
  average := (x + y + z) / 3.0;  
}
```

Total Token Usage

Input tokens: 282

Output tokens: 63

Reasoning tokens: 26

Sum of ‘total tokens’: 345

Experiment Timings

Overall Experiment started at 1780564041692, ended at 1780564076798, lasting 35106ms (35.11 seconds)

Iteration #1 started at 1780564041797, ended at 1780564076795, lasting 34998ms (35.00 seconds)